

Indian Institute of Engineering Science and Technology, Shibpur
Department of Computer Science and Technology

Algorithm Lab (CS2271)
Assignment 2

1. Write a C program that dynamically creates an array of integers of size n , where n is a positive integer such that $n \geq 10$. The value of n must be read from the user at runtime. Populate the array with randomly generated integer values. After generating the array, design and implement an efficient algorithm to determine the **4th largest** and the **3rd smallest** elements present in the array. The solution should avoid sorting the entire array and must operate in **linear time complexity, $O(n)$** . Finally, display the generated array along with the computed 4th largest and 3rd smallest elements as output.

2. In this assignment, you are required to develop **two separate programs** to determine the **maximum** and **minimum** elements in an array of n integers. The first program should implement the **naïve Max–Min algorithm**, which finds the maximum and minimum values by scanning the array sequentially. The second program should implement the **Max–Min algorithm using the Divide and Conquer strategy**, where the array is recursively divided into smaller subarrays and the maximum and minimum elements are determined efficiently. Execute both programs on **randomly generated input arrays** of varying sizes, specifically: $n=100, 1000, 10,000, 20,000, \dots, 100,000$. For each input size, **count and record the number of element comparisons** performed by each algorithm. Finally, plot a graph with the **input size (n)** on the **X-axis** and the **number of comparisons** on the **Y-axis** to visually compare the performance of the two approaches.

3. Consider a source string $S[0 \dots n-1]$ of length n , composed exclusively of the symbols 'c' and 'd'. You are also given a pattern string $P[0 \dots m-1]$ of length m , where $m \ll n$. The pattern string consists of the symbols 'c', 'd', and the special character '*'. The symbol '*' acts as a wildcard and can match **any single character**, either 'c' or 'd', in the source string. All other symbols in the pattern must match the corresponding characters in the source string exactly. The objective is to identify all **valid match positions** in the source string where the pattern occurs. A position j is considered a valid match if the pattern P matches the substring $S[j \dots j+m-1]$. The output should be a **sorted list M** containing all such valid starting indices j in the source string.

Example: If the source string is $S = c d c d d c d$ and the pattern is $P = c d *$, then the pattern matches the source string starting at positions **0** and **2**. Therefore, the output list M should be **[0, 2]**.

4. Design an algorithm to create two sets of integers, namely **A** and **B**. The number of elements in each set must be entered by the user at runtime. While constructing the sets, ensure that each set contains **only distinct elements**, thereby preventing the insertion of duplicate values. After successfully creating the two sets, develop a procedure to determine whether **set A is a subset of set B**. Additionally, verify whether **A is a proper subset of B**.

5. You are given **k disjoint or overlapping sets** $S_1, S_2, \dots, S_k$ each containing up to **n** integers. Design an algorithm to compute the **union** of all **k** sets.